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Department : CSE

Subject : Object Oriented Programming Language

Batch : 04

Project -1-Student INFO Manager

AIM:

To develop a Java program using Object Oriented Programming concepts to store and display details of multiple students such as name, ID, department, age, and college name.

PROCEDURE:

1. Open Notepad or Visual Studio Code.
2. Create a new file and save it as Main.java.
3. Import required packages using `import java.util.*;`
4. Create a Student class.
5. Declare data members such as name, id, department, age, and a static college name.
6. Define a parameterized constructor to initialize student details.
7. Create a `display()` method to print the student information.
8. Create the Main class with the `main()` method.
9. Use the Scanner class to get input from the user.
10. Read the number of students.
11. Create an array of Student objects.
12. Use a loop to read student details and create objects.
13. Store each object in the array.
14. Use another loop to call the `display()` method for each student.
15. Save the program file.
16. Compile the program using `javac Main.java`.
17. Run the program using `java Main`.

18. Verify that the student details are displayed correctly. CODE:

CODE:

```
import java.util.*;
import java.io.*;
class Student {
    String name;
    int id;
    String department;
    int age;
    static String collegeName = "Prince Engineering College";
    Student() {}
    Student(String name, int id, String department, int age) {
        this.name = name;
        this.id = id;
        this.department = department;
        this.age = age;
    }
    void display() {
        System.out.println("Name      : " + name);
        System.out.println("ID        : " + id);
        System.out.println("College   : " + collegeName);
        System.out.println("Department : " + department);
        System.out.println("Age       : " + age);
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter number of students: ");

int n = sc.nextInt();

Student[] students = new Student[n];

for (int i = 0; i < n; i++) {

    System.out.println("\nEnter details for Student " + (i + 1));

    System.out.print("Name: ");

    sc.nextLine();

    String name = sc.nextLine();

    System.out.print("ID: ");

    int id = sc.nextInt();

    System.out.print("Department: ");

    sc.nextLine();

    String dept = sc.nextLine();

    System.out.print("Age: ");

    int age = sc.nextInt();

    students[i] = new Student(name, id, dept, age);

}

for (int i = 0; i < n; i++) {

    students[i].display();

}

}
```

```
Enter number of students: 1

Enter details for Student 1
Name: Jai Surya
ID: 505
Department: CSE
Age: 18
Name      : Jai Surya
ID        : 505
College   : Prince Engineering College
Department : CSE
Age       : 18

=== Code Execution Successful ===
```

RESULT:

Thus, the Java program was successfully executed using OOPS concepts such as class, Object, Constructor, static variable, and array of objects to store and display Student Details